

Section 1: Negative and Rational Exponents

1) Evaluate. Express as a fraction in lowest terms.

a) 10^{-1}

$$= \frac{1}{10}$$

b) 4^{-2}

$$= \frac{1}{4^2}$$

$$= \frac{1}{16}$$

c) $3^{-2} + 9^{-1}$

$$= \frac{1}{9} + \frac{1}{9}$$

$$= \frac{2}{9}$$

d) $5^{-3} + 5^0$

$$= \frac{1}{125} + 1$$

$$= \frac{126}{125}$$

e) $\left(\frac{1}{5}\right)^{-1}$

$$= \left(\frac{5}{1}\right)^1$$

$$= 5$$

f) $\left(\frac{3}{4}\right)^{-3}$

$$= \left(\frac{4}{3}\right)^3$$

$$= \frac{64}{27}$$

2) Simplify. Express your answers using only positive exponents.

a) $(x^{-2})(x^{-1})(x^0)$

$$= x^{-3}$$

$$= \frac{1}{x^3}$$

b) $(3km^2)(2k^{-2}m^{-2})$

$$= 6k^{-1}m^0$$

$$= \frac{6}{k}$$

c) $w^{-3} \div w^{-2}$

$$= w^{-1}$$

$$= \frac{1}{w}$$

d) $\frac{u^{-2}v^3}{u^{-3}v^{-2}}$

$$= uv^5$$

e) $(z^{-3})^{-2}$

$$= z^6$$

f) $(2ab^{-1})^{-2}$

$$= \frac{1}{(2ab^{-1})^2}$$

$$= \frac{1}{4a^2b^{-2}}$$

$$= \frac{b^2}{4a^2}$$

3) Simplify. Express your answers using only positive exponents.

a) $(4a^{-2})(-2a^{-3})$

$$= -8a^{-5}$$

$$= \frac{-8}{a^5}$$

b) $\frac{(2x^2y)^{-2}(3xy)^{-1}}{(6x^2y^2)^{-2}}$

$$= \frac{(6x^2y^2)^2}{(2x^2y)^2(3xy)^1}$$

$$= \frac{36x^4y^4}{4x^4y^2(3xy)}$$

$$= \frac{36x^4y^4}{12x^5y^3}$$

$$= 3x^{-1}y$$

$$= \frac{3y}{x}$$

c) $\left(\frac{1}{4x^2}\right)^{-2}$

$$= \left(\frac{4x^2}{1}\right)^2$$

$$= 16x^4$$

d) $\left(\frac{6a^3}{4b^4}\right)^{-2}$

$$= \left(\frac{4b^4}{6a^3}\right)^2$$

$$= \frac{16b^8}{36a^6}$$

$$= \frac{4b^8}{9a^6}$$

4) Evaluate.

a) $\sqrt[3]{64}$

$$= 4$$

b) $\sqrt[4]{625}$

$$= 5$$

c) $\sqrt[5]{-3125}$

$$= -5$$

d) $\left(\frac{1}{64}\right)^{\frac{1}{6}}$

$$= \frac{1}{2}$$

e) $27^{\frac{2}{3}}$

$$= \left(27^{\frac{1}{3}}\right)^2$$

$$= (3)^2$$

$$= 9$$

f) $(-1000)^{\frac{4}{3}}$

$$= \left[(-1000)^{\frac{1}{3}}\right]^4$$

$$= (-10)^4$$

$$= 10000$$

g) -4^{-3}

$$= \frac{-1}{4^3}$$

$$= \frac{-1}{64}$$

h) $\left(\frac{3}{4}\right)^{-2}$

$$= \left(\frac{4}{3}\right)^2$$

$$= \frac{16}{9}$$

i) $\left(-\frac{27}{125}\right)^{-\frac{2}{3}}$

$$= \left[\left(-\frac{125}{27}\right)^{\frac{1}{3}}\right]^2$$

$$= \left(-\frac{5}{3}\right)^2$$

$$= \frac{25}{9}$$

5) Simplify. Express answers using only positive exponents.

a) $n^{\frac{1}{2}} \times n^{\frac{1}{3}} \times n^{\frac{1}{4}}$

$$= n^{6/12} \cdot n^{4/12} \cdot n^{3/12}$$

$$= n^{13/12}$$

b) $(27y^3)^{\frac{1}{3}} \times \left(\frac{1}{16y^4}\right)^{-\frac{3}{4}}$

$$= 3y \times \left(\frac{16y^4}{1}\right)^{\frac{3}{4}}$$

$$= 3y(8y^3)$$

$$= 24y^4$$

c) $(27x^6)^{\frac{2}{3}} \div (9x^4)^{\frac{1}{2}}$

$$= 9x^4 \div 3x^2$$

$$= 3x^2$$

d) $\frac{x^{-\frac{2}{3}}}{x^{-\frac{4}{5}}}$

$$= \frac{x^{4/5}}{x^{2/3}}$$

$$= \frac{x^{12/15}}{x^{10/15}}$$

$$= x^{2/15}$$

Section 2: Rational Expressions

6) Simplify each expression and state all restrictions on x .

a) $\frac{x+7}{x^2+10x+21}$

$$= \frac{\cancel{x+7}}{(x+7)(x+3)}$$

$$= \frac{1}{x+3}, x \neq -7, -3$$

b) $\frac{x^2-64}{x-8}$

$$= \frac{(x-8)(x+8)}{\cancel{x-8}}$$

$$= x+8, x \neq 8$$

c) $\frac{x^2-9}{x^2-8x+15}$

$$= \frac{(\cancel{x-3})(x+3)}{(\cancel{x-3})(x-5)}$$

$$= \frac{x+3}{x-5}, x \neq 3, 5$$

7) Simplify each expression and state the restrictions.

a) $\frac{3x^2}{5xy} \times \frac{20xy^3}{12xy}$

$$= \frac{60x^3y^3}{60x^2y^2}$$

$$= xy, x \neq 0, y \neq 0$$

b) $\frac{150a^3b^4}{20a^2b} \div \frac{6b}{8ab^2}$

$$= \frac{150a^3b^4}{20a^2b} \times \frac{8ab^2}{6b}$$

$$= \frac{60a^4b^6}{6a^2b^2}$$

$$= 10a^2b^4, a \neq 0, b \neq 0$$

$$c) \frac{1}{3x} + \frac{5}{2x^2}$$

$$= \frac{2x(1)}{2x(3x)} + \frac{3(5)}{3(2x^2)}$$

$$= \frac{2x}{6x^2} + \frac{15}{6x^2}$$

$$= \frac{2x+15}{6x^2} ; x \neq 0$$

8) Simplify each expression and state restrictions.

$$a) \frac{x^2+7x}{3x+21} \times \frac{x^2+3x+2}{x+2}$$

$$= \frac{x(x+7)}{3(x+7)} \cdot \frac{(x+2)(x+1)}{(x+2)}$$

$$= \frac{x(x+1)}{3} ; x \neq -7, -2$$

$$d) \frac{4}{x-6} - \frac{3}{x-4}$$

$$= \frac{4(x-4)}{(x-6)(x-4)} - \frac{3(x-6)}{(x-6)(x-4)}$$

$$= \frac{4x-16-3x+18}{(x-6)(x-4)}$$

$$= \frac{x+2}{(x-6)(x-4)} ; x \neq 6, 4$$

$$b) \frac{x^2+4x-60}{3x+30} \div \frac{x^2-8x+12}{6x-12}$$

$$= \frac{(x+10)(x-6)}{3(x+10)} \times \frac{6(x-2)}{(x-6)(x-2)}$$

$$= \frac{6}{3}$$

$$= 2 ; x \neq -10, 2, 6$$

$$c) \frac{3}{x^2+7x+10} - \frac{5x}{x^2-4}$$

$$= \frac{3}{(x+2)(x+5)} - \frac{5x}{(x-2)(x+2)}$$

$$= \frac{3(x-2)}{(x+2)(x+5)(x-2)} - \frac{5x(x+5)}{(x+2)(x+5)(x-2)}$$

$$= \frac{3x-6-5x^2-25x}{(x+2)(x+5)(x-2)}$$

$$= \frac{-5x^2-22x-6}{(x+2)(x+5)(x-2)} ; x \neq -5, -2, 2$$

$$d) \frac{-10x}{x^2+18x+32} + \frac{12x}{x^2+6x-160}$$

$$= \frac{-10x}{(x+16)(x+2)} + \frac{12x}{(x+16)(x-10)}$$

$$= \frac{-10x(x+10)}{(x+16)(x+2)(x-10)} + \frac{12x(x+2)}{(x+16)(x+2)(x-10)}$$

$$= \frac{-10x^2+100x+12x^2+24x}{(x+16)(x+2)(x-10)}$$

$$= \frac{2x^2+124x}{(x+16)(x+2)(x-10)}$$

$$= \frac{2x(x+62)}{(x+16)(x+2)(x-10)} ; x \neq -16, -2, 10$$

9) Simplify each expression and state any restrictions

$$a) \frac{x-8}{x+7} \times \frac{x+15}{x^2+12x-45}$$

$$= \frac{x-8}{x+7} \cdot \frac{x+15}{(x+15)(x-3)}$$

$$= \frac{x-8}{(x+7)(x-3)} ; x \neq -7, 3$$

$$b) \frac{x^2+12x+20}{x+5} \div \frac{x^2+7x-30}{x+10}$$

$$= \frac{(x+10)(x+2)}{x+5} \cdot \frac{x+10}{(x+10)(x-3)}$$

$$= \frac{(x+10)(x+2)}{(x+5)(x-3)} ; x \neq -10, -5, 3$$

$$c) \frac{x+3}{x-7} - \frac{x+9}{x-2}$$

$$= \frac{(x+3)(x-2)}{(x-7)(x-2)} - \frac{(x+9)(x-7)}{(x-7)(x-2)}$$

$$= \frac{x^2+x-6 - (x^2+2x-63)}{(x-7)(x-2)}$$

$$= \frac{-1x+57}{(x-7)(x-2)} ; x \neq 2, 7$$

$$e) \frac{5x+1}{2x-1} - \frac{3x-3}{1-2x}$$

$$= \frac{-1(5x+1)}{-1(2x-1)} - \frac{3x-3}{1-2x}$$

$$= \frac{-5x-1}{-2x+1} - \frac{3x-3}{1-2x}$$

$$= \frac{-5x-1}{1-2x} - \frac{3x-3}{1-2x}$$

$$= \frac{-5x-1-(3x-3)}{1-2x}$$

$$= \frac{-8x+2}{1-2x}$$

$$\frac{-2(4x-1)}{1-2x} ; x \neq \frac{1}{2}$$

$$d) \frac{x+8}{x+3} + \frac{x-6}{x^2+9x+18}$$

$$= \frac{x+8}{x+3} + \frac{x-6}{(x+3)(x+6)}$$

$$= \frac{(x+6)(x+8)}{(x+3)(x+6)} + \frac{x-6}{(x+3)(x+6)}$$

$$= \frac{x^2+14x+48+x-6}{(x+3)(x+6)}$$

$$= \frac{x^2+15x+42}{(x+3)(x+6)} ; x \neq -3, -6$$

Answers

1) a) $\frac{1}{10}$ b) $\frac{1}{16}$ c) $\frac{2}{9}$ d) $\frac{126}{125}$ e) 5 f) $\frac{64}{27}$

2) a) $\frac{1}{x^3}$ b) $\frac{6}{k}$ c) $\frac{1}{w}$ d) uv^5 e) z^6 f) $\frac{b^2}{4a^2}$

3) a) $-\frac{8}{a^5}$ b) $\frac{3y}{x}$ c) $16x^4$ d) $\frac{4b^8}{9a^6}$

4) a) 4 b) 5 c) -5 d) $\frac{1}{2}$ e) 9 f) 10 000 g) $-\frac{1}{64}$ h) $\frac{16}{9}$ i) $\frac{25}{9}$

5) a) $n^{\frac{13}{12}}$ b) $24y^4$ c) $3x^2$ d) $x^{\frac{2}{15}}$

6) a) $\frac{1}{x+3}, x \neq -7, x \neq -3$ b) $x+8, x \neq 8$ c) $\frac{x+3}{x-5}, x \neq 3, 5$

7) a) $xy, x \neq 0, y \neq 0$ b) $10a^2b^4, a \neq 0, b \neq 0$ c) $\frac{2x+15}{6x^2}, x \neq 0$ d) $\frac{x+2}{(x-4)(x-6)}, x \neq 4, x \neq 6$

8) a) $\frac{x(x+1)}{3}, x \neq -7, -2$ b) $2, x \neq -10, 2, 6$ c) $\frac{-5x^2-22x-6}{(x+5)(x+2)(x-2)}, x \neq -5, -2, 2$

d) $\frac{2x(x+62)}{(x+16)(x+2)(x-10)}, x \neq -16, -2, 10$

9) a) $\frac{x-8}{(x+7)(x-3)}, x \neq -15, -7, 3$ b) $\frac{(x+10)(x+2)}{(x+5)(x-3)}, x \neq -10, -5, 3$ c) $\frac{-x+57}{(x-7)(x-2)}, x \neq 2, 7$

d) $\frac{x^2+15x+42}{(x+6)(x+3)}, x \neq -6, -3$ e) $\frac{8x-2}{2x-1}, x \neq \frac{1}{2}$